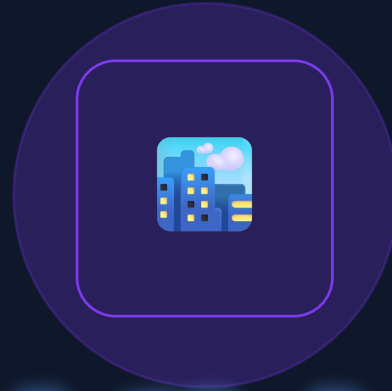


CLUJ AI BUILDATHON 2026



Cluj Civic AI

AI-powered urban infrastructure reporting
for a smarter Cluj-Napoca

Claude Vision AI

Next.js 15

PostgreSQL

Real-time Maps

Citizen & Gov Panels

Built at Cluj AI Buildathon · May 2026

THE PROBLEM

Urban infrastructure issues go unreported & unresolved

Citizens encounter daily problems — but reporting is slow, fragmented, and produces no visible action.



No easy channel

Hotlines, social posts — no systematic follow-up



Manual categorisation

Every report triaged by hand — delays & misrouting



No location tracking

Problems lack GPS context for dispatch



Zero transparency

Citizens never know if their report was acted on



No data for city hall

Authorities can't plan maintenance budgets



Critical issues missed

Hazards stay unresolved for weeks

One app. AI at the core. City-wide impact.

Cluj Civic AI connects citizens to local government through an intelligent, photo-first reporting platform.

1

Snap a photo

Citizen photographs a problem — pothole, broken light, water leak — with their phone.

2

Claude Vision analyses instantly

AI classifies issue type, severity (LOW→CRITICAL), and writes a human-readable summary in seconds.

3

GPS-pinned & submitted

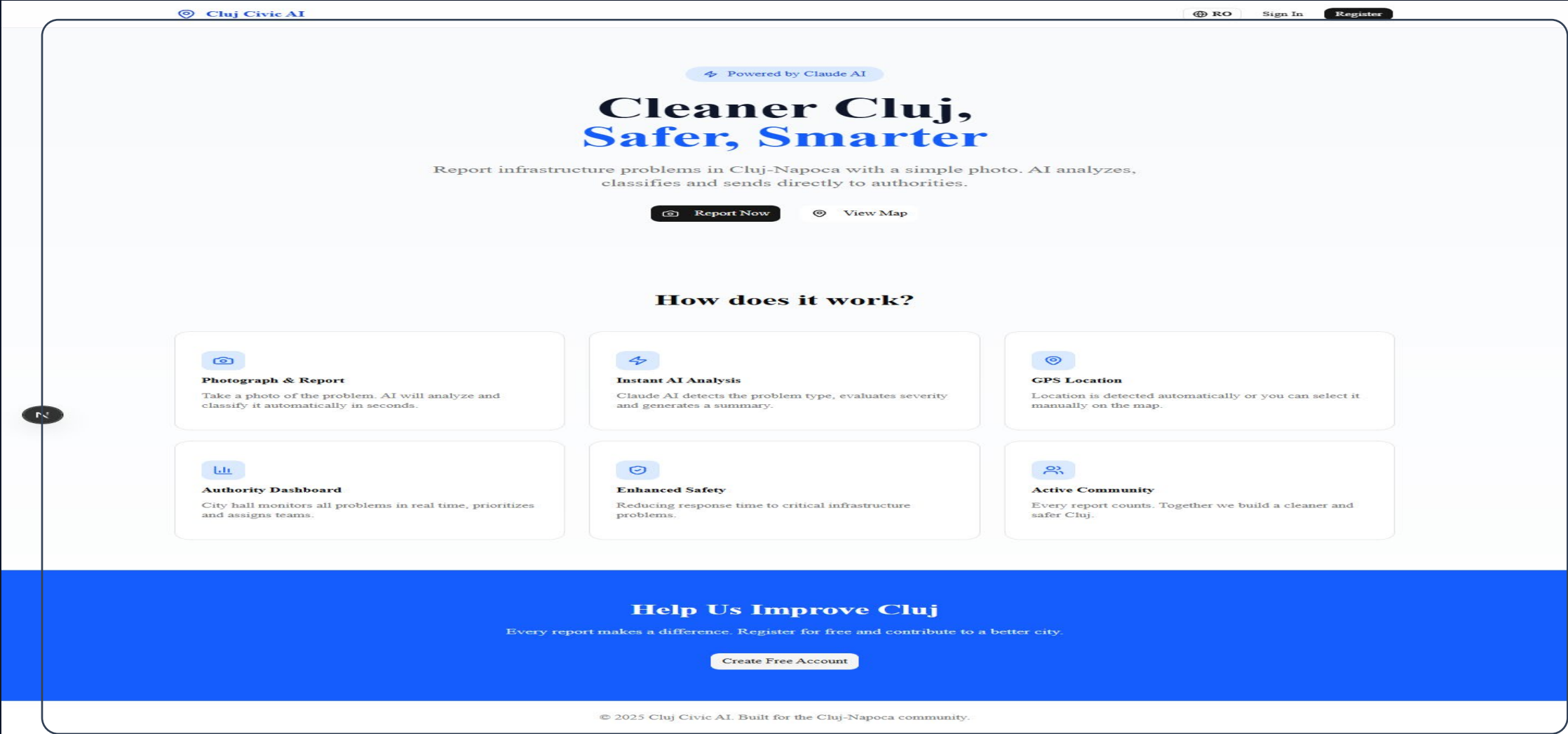
The report is geo-tagged and routed to government officers with full context already attached.

4

Track resolution in real time

Citizens follow status updates; officers log actions; the city gets aggregate analytics.

A clean, modern citizen-facing homepage



Log in Page



Cluj Civic AI

Sign in to the smart city platform

DEMO ACCOUNTS



Citizen



Gov Officer



Admin

OR SIGN IN MANUALLY

Email

Password

[Forgot password?](#)

Sign In

Don't have an account? [Register](#)

Personal dashboard — at a glance

Hello, KHAN!

Track your infrastructure reports

 **3**
Total Reports

 **3**
Pending

 **0**
In Progress

 **0**
Resolved

 Report Problem

Recent Reports

[View All](#)

 Broken Street Light

Under Review

High

📍 Parcul Central, Center, Cluj-Napoca, Cluj Metropolitan Area, ...

📅 38 minutes ago



 Illegal Garbage

Under Review

High

📍 Strada Costache Negruzzi, Gruia, Cluj-Napoca, Cluj Metropoli...

 about 1 hour ago

 Road Crack

Pending

Medium

36, Strada Între Lacuri, Între Lacuri, Cluj-Napoca, Cluj Metro...

 about 18 hours ago

4-step guided report submission

Cluj Civic AI

Dashboard

Report Problem

My Reports

Map

Transparency

Report a Problem

Take a photo and AI will automatically classify the issue

1 Image — 2 Location — 3 Details — 4 Submit

Step 1: Upload an image

Drag an image or click to select
PNG, JPG, WebP up to 10MB

Select image

< Back

Continue >



Upload photo

Drag-drop or camera. Claude Vision analyses it instantly.



Pin location

GPS auto-detects or click the map to place a pin.



Review details

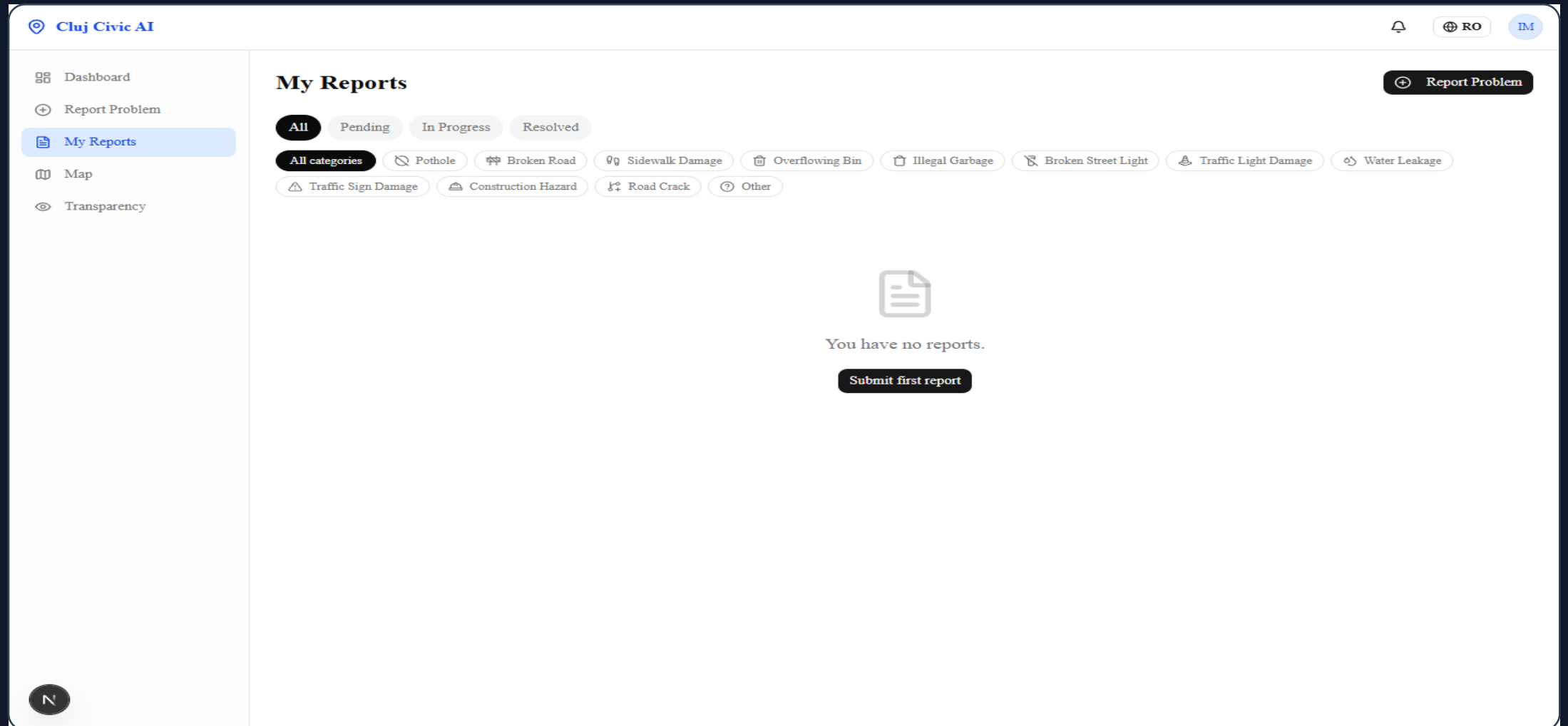
AI pre-fills type, severity & summary — citizen can edit.



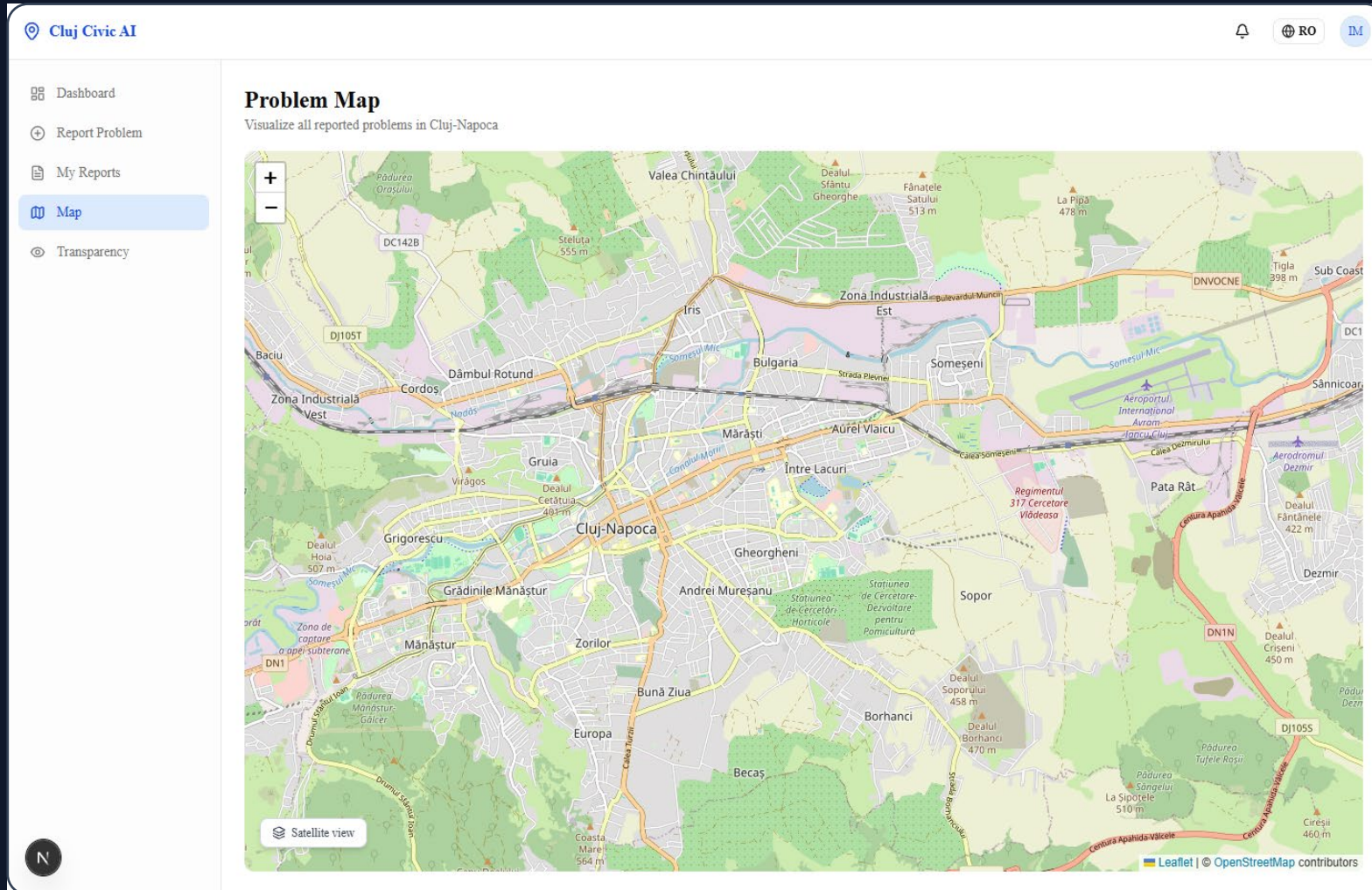
Submit

One tap. Report goes live in the government dashboard.

Filter by status + category — track every report



All reports on the map — street & satellite view



GPS-pinned



Every report carries exact lat/long coordinates.

Satellite layer



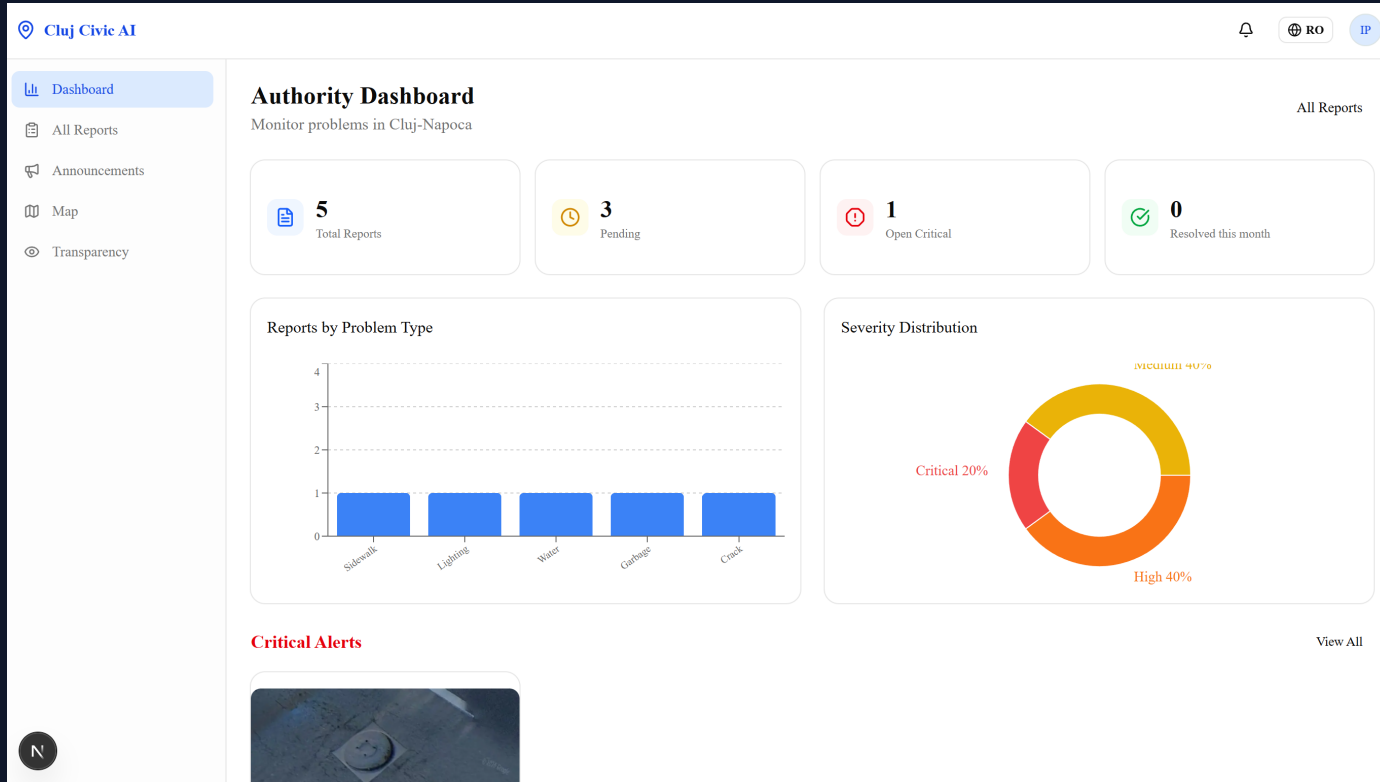
Toggle between street map and Esri satellite imagery.

Severity colours



Markers coloured LOW→CRITICAL for instant triage.

Officers see everything — analytics, reports, actions



Charts

Issue type bar chart + severity pie chart.



All reports

Filter, sort, update status, log officer actions.



Status updates

Each status change is visible to the citizen instantly.

Officers see everything — analytics, reports, actions

📍 Cluj Civic AI

🔔

🌐 RO

IP

🏠 Dashboard

📋 All Reports

📢 Announcements

🗺 Map

👁 Transparency

All Reports (5)

All

Critical

Pending

Accepted

Work Started

In Progress

Delayed

Completed

Img	Type	Severity	Status	Citizen	Location	Date	Action
	Broken Street Light	High	Under Review	KHAN MD IMRAN	Parcul Central, Center, Cluj-N...	about 2 hours ago	Manage
	Illegal Garbage	High	Under Review	KHAN MD IMRAN	Strada Costache Negruzzi, Gru...	about 2 hours ago	Manage
	Road Crack	Medium	Pending	KHAN MD IMRAN	36, Strada Între Lacuri, Între L...	about 19 hours ago	Manage
	Sidewalk Damage	Medium	Pending	Admin Cluj	40, Strada Ciocârliei, Mărăști, ...	3 days ago	Manage
	Water Leakage	Critical	Pending	KHAN MD IMRAN	Cloister of the Monastery of th...	4 days ago	Manage

Officers see everything — analytics, reports, actions

 Cluj Civic AI



 Dashboard

 All Reports

 Announcements

 Map

 Transparency



Logged in as Officer

←

Road Crack

ID: cmpigymnj0001nkvnsosxqvho



Medium

Pending

 36, Strada Între Lacuri, Între Lacuri, Cluj-Napoca, Cluj Metropolitan Area, Cluj, 400591, Romania

 KHAN MD IMRAN (misukkhanimranwd@gmail.com)

 23 May 2026, 17:54

AI Analysis

Transverse and longitudinal cracks on the asphalt surface indicating early structural deterioration.

Citizen Description

Transverse and longitudinal cracks on the asphalt surface indicating early structural deterioration.

Action History

 Raport trimis

about 19 hours ago

Update Report

New Status

PENDING

▼

Assigned Team (optional)

e.g. Road Maintenance Team 2

Action Taken *

Describe what was done or what will be done...

☒ Visible to public

Update Report

Powered by Claude Vision API

claude-sonnet-4-6 → structured JSON output → auto-filled form

```
{  
  
  "issueType":      "POTHOLE",  
  
  "severity":       "HIGH",  
  
  "summary":        "Deep pothole, risk of vehicle damage",  
  
  "confidence":     91,  
  
  "additionalNotes": "Temporary marking recommended",  
  
}
```

12 Issue Types

POTHOLE · BROKEN_ROAD · SIDEWALK_DAMAGE · GARBAGE · WATER_LEAKAGE ·
TRAFFIC_LIGHT_DAMAGE · and 6 more

4 Severity Levels

LOW cosmetic → MEDIUM weeks → HIGH hazard → CRITICAL same-day

Demo Fallback

Hash-based deterministic scenarios ensure the app never breaks if AI is unavailable.

Production-grade full-stack architecture



Next.js 15

App Router + TypeScript



Claude Sonnet

Vision AI Analysis



NextAuth v5

JWT Auth + Roles



PostgreSQL

Prisma ORM



Cloudinary

Image Storage & CDN



Leaflet + OSM

Interactive Maps



Recharts

Analytics Charts



Nodemailer

Password Reset Emails



Tailwind + shadcn

UI Components



Zod + RHF

Validation & Forms



i18n Ready

Multi-language Support



pm2

Persistent Server

Roadmap to a full city platform

✓ Phase 1 — MVP (Done)

- Citizen photo reporting
- Claude Vision AI classification
- Government dashboard
- GPS maps (Street + Satellite)
- Report editing & deletion
- Forgot password + email
- Account management
- Category & status filtering
- Analytics charts

→ SOON Phase 2 — Q3 2026

- Push / SMS notifications
- Mobile PWA (installable)
- Duplicate report detection
- Heat-map density view
- Officer assignment & teams
- Public report feed
- Multi-language (RO / HU / EN)

🚀 Phase 3 — 2027

- Predictive maintenance AI
- Integration with Cluj City API
- Resident satisfaction surveys
- Budget allocation AI
- Open data export
- Multi-city rollout (Romania)

Thank you!

Let's build a smarter Cluj.

Cluj Civic AI proves that AI + civic tech can meaningfully improve quality of life — one photo at a time.

 **Live Demo**

 **Submit Report**

 **View Map**



Claude Sonnet 4.6

Anthropic Vision API



Next.js 15 + TypeScript

Production-ready stack



Open Infrastructure

OpenStreetMap + OSS